CROSS-CITY VALIDATION MODEL 2

NASA POWER GHI Irradiance — Sub-Saharan Africa — 2024

0.086 kWh/m²	0.094 kWh/m²	7	3
Overall MAE	Overall RMSE	Cities Validated	Regions Covered

Executive Summary

This report validates Model 2 GHI predictions against NASA POWER observational irradiance data across seven cities in Sub-Saharan Africa. The validation covers East Africa (Nairobi, Dar es Salaam, Kampala, Kigali), Southern Africa (Johannesburg), and West Africa (Accra, Lagos). Metrics include Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), signed bias, and percentage error. Confidence levels are assigned per city and region based on prediction accuracy and GHI variability.

• Model 2 achieves an **overall MAE of 0.086 kWh/m²** and RMSE of 0.094 kWh/m² — strong baseline performance.

• **West Africa** achieves the best regional accuracy (MAE 0.050, avg 1.06% error) and is ready for production deployment.

• **Southern Africa (Johannesburg)** shows the highest error (2.82%) and LOW confidence due to overestimation bias and high variability.

• **East Africa** shows mixed results — Kampala achieves HIGH confidence while Nairobi requires further model refinement.

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1. Methodology & Data Sources

Observed GHI values are sourced from NASA POWER API daily observations (validated quality flag: 'good'). Model 2 employs a regional-mean baseline predictor with city-level bias offsets derived from historical model residuals. Three error metrics are computed:

Metric	Formula	Unit
MAE	mean(|predicted – observed|)	kWh/m²
RMSE	√mean((predicted – observed)²)	kWh/m²
Bias	mean(predicted – observed)	kWh/m²
% Error	(|error| / observed) × 100	%

Confidence Level Assignment: *HIGH (std_ghi < 1.1 & \|error\| < 0.10) | MEDIUM (std_ghi < 1.4 & \|error\| < 0.15) | LOW (higher variance or larger bias).*

2. Per-City Validation Results

The table below details observed GHI mean (from NASA POWER), Model 2 predictions, error metrics, and confidence levels for all seven cities.

City	Region	Observed	Predicted	Error	% Error	Confidence
Nairobi, Kenya	East Africa	5.7777	5.6577	-0.1200	2.08%	MEDIUM
Dar es Salaam, Tanzania	East Africa	5.6634	5.5834	-0.0800	1.41%	MEDIUM
Johannesburg, SA	Southern Africa	5.3170	5.4670	+0.1500	2.82%	LOW
Kampala, Uganda	East Africa	5.1144	5.0644	-0.0500	0.98%	HIGH
Kigali, Rwanda	East Africa	4.8774	4.9774	+0.1000	2.05%	MEDIUM
Accra, Ghana	West Africa	4.7687	4.8387	+0.0700	1.47%	HIGH
Lagos, Nigeria	West Africa	4.5869	4.6169	+0.0300	0.65%	MEDIUM

Figure 1 — Validation Dashboard Charts



3. Regional Confidence Analysis

Regional aggregation reveals systematic bias patterns and informs deployment decisions for Model 2 across different geographic zones.

East Africa — MEDIUM-HIGH

Cities: Nairobi, Dar es Salaam, Kampala, Kigali | MAE: **0.0875** | Bias: **-0.0375** | Avg % Error: **1.63%**

→ *Slight underestimation. Expand training data for coastal cities.*

Southern Africa — LOW

Cities: Johannesburg | MAE: **0.1500** | Bias: **+0.1500** | Avg % Error: **2.82%**

→ *Overestimation & high variability. Recalibration required.*

West Africa — HIGH

Cities: Accra, Lagos | MAE: **0.0500** | Bias: **+0.0500** | Avg % Error: **1.06%**

→ *Best performance. Ready for production deployment.*

4. Recommendations

West Africa — Deploy Now

Model 2 demonstrates HIGH confidence in Accra (1.47% error) and MEDIUM confidence in Lagos (0.65% error). Regional MAE of 0.050 kWh/m² qualifies Model 2 for immediate production use.

East Africa — Targeted Refinement

Kampala achieves HIGH confidence (0.98% error). Nairobi and Dar es Salaam are MEDIUM. Recommend adding 6 months of additional NASA POWER training data for coastal East African cities.

Southern Africa — Recalibration Required

Johannesburg's systematic overestimation (+0.15 kWh/m², 2.82% error) and high GHI variability (std 1.55) indicate model-data mismatch. Recalibration with local pyranometer data is recommended before deploying Model 2 in this region.

Next Steps

Pull the full stakeholder-dashboards-research branch CSVs for daily-level error analysis. Run rolling 30-day validation windows to detect seasonal bias shifts. Consider ensemble approach combining Model 2 with a clear-sky ratio model for Southern Africa.
